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Q1. Memory saving**CODE**

```
#include <stdio.h>

union SensorReading {
    float heartRate;
    int stepCount;
    float sleepQuality;
};

struct SensorReadingStruct {
    float heartRate;
    int stepCount;
    float sleepQuality;
};

int main() {
    union SensorReading reading;
    struct SensorReadingStruct readingStruct;

    reading.heartRate = 72.5;
    printf("Union size: %lu bytes\n", sizeof(reading));
    printf("Structure size: %lu bytes\n", sizeof(readingStruct));

    printf("\nSince only one reading is transmitted at a time,\n");
    printf("a union is used to save memory by sharing the same space.\n");

    return 0;
}
```

INPUT

none

OUTPUT

Q2. Structure pointer**CODE**

```
#include <stdio.h>
#include <string.h>

union ClaimDetails {
    char hospitalDetails[50];
    char accidentDetails[50];
    char propertyDetails[50];
};

struct Claim {
    int claimID;
    char policyholderName[50];
    char claimDate[15];
    char status[20];
    union ClaimDetails details;
};

void updateStatus(struct Claim *ptr, char *newStatus) {
    strcpy(ptr->status, newStatus);
}

int main() {
    struct Claim myClaim;
    myClaim.claimID = 101;
    strcpy(myClaim.policyholderName, "Amrutha Yazhini");
    strcpy(myClaim.claimDate, "2023-10-27");
    strcpy(myClaim.status, "Pending");

    strcpy(myClaim.details.hospitalDetails, "City General Hospital");

    printf("--- Initial Claim Info ---\n");
    printf("ID: %d\n", myClaim.claimID);
    printf("Name: %s\n", myClaim.policyholderName);
    printf("Status: %s\n", myClaim.status);
    printf("Detail: %s\n", myClaim.details.hospitalDetails);

    updateStatus(&myClaim, "Approved");

    printf("\n--- Updated Claim Info ---\n");
    printf("New Status: %s\n", myClaim.status);

    return 0;
}
```

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INPUT

```
101
Amrutha Yazhini
City Hospital
Pending
Approved
```

OUTPUT

Q3. Food delivery**CODE**

```
#include <stdio.h>
#include <string.h>

struct Customer {
    char name[50];
    char phone[15];
};

struct Address {
    char street[50];
    char city[30];
};

union Payment {
    float wallet_balance;
    char card_number[20];
    char cash_on_delivery_mode[10];
};

struct Order {
    int order_id;
    struct Customer customer;
    struct Address address;
    int payment_type;
    union Payment payment;
};

int main() {
    struct Order myOrder;

    myOrder.order_id = 101;

    strcpy(myOrder.customer.name, "Amrutha");
    strcpy(myOrder.customer.phone, "9876543210");

    strcpy(myOrder.address.street, "Main Street");
    strcpy(myOrder.address.city, "Chennai");

    myOrder.payment_type = 1;
    myOrder.payment.wallet_balance = 500.50;

    printf("--- Order Details ---\n");
    printf("Order ID: %d\n", myOrder.order_id);
```

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```
printf("Customer: %s (%s)\n", myOrder.customer.name, myOrder.customer.phone);
printf("Address: %s, %s\n", myOrder.address.street, myOrder.address.city);

if (myOrder.payment_type == 1) {
    printf("Payment Method: Wallet\n");
    printf("Amount: %.2f\n", myOrder.payment.wallet_balance);
} else if (myOrder.payment_type == 2) {
    printf("Payment Method: Card\n");
    printf("Card No: %s\n", myOrder.payment.card_number);
} else {
    printf("Payment Method: Cash on Delivery\n");
}

return 0;
}
```

INPUT

1

OUTPUT

Q4. Digital wallet application**CODE**

```
#include <stdio.h>
#include <string.h>

union Currency {
    float standard;
    double crypto;
    int rewards;
};

struct Transaction {
    int transactionID;
    char userID[20];
    long timestamp;
    union Currency amount;
    char type;
};

int main() {
    struct Transaction t1;

    t1.transactionID = 101;
    strcpy(t1.userID, "user_01");
    t1.timestamp = 1672531200;
    t1.type = 'S';
    t1.amount.standard = 50.75;

    printf("Transaction ID: %d\n", t1.transactionID);
    printf("User ID: %s\n", t1.userID);

    if (t1.type == 'S') {
        printf("Amount: %.2f (Standard Currency)\n", t1.amount.standard);
    } else if (t1.type == 'C') {
        printf("Amount: %lf (Crypto)\n", t1.amount.crypto);
    } else if (t1.type == 'R') {
        printf("Amount: %d (Rewards)\n", t1.amount.rewards);
    }

    printf("\n--- Memory Analysis ---\n");
    printf("Size of union Currency: %zu bytes\n", sizeof(union Currency));
    printf("Size of struct Transaction: %zu bytes\n", sizeof(struct Transaction));

    return 0;
}
```